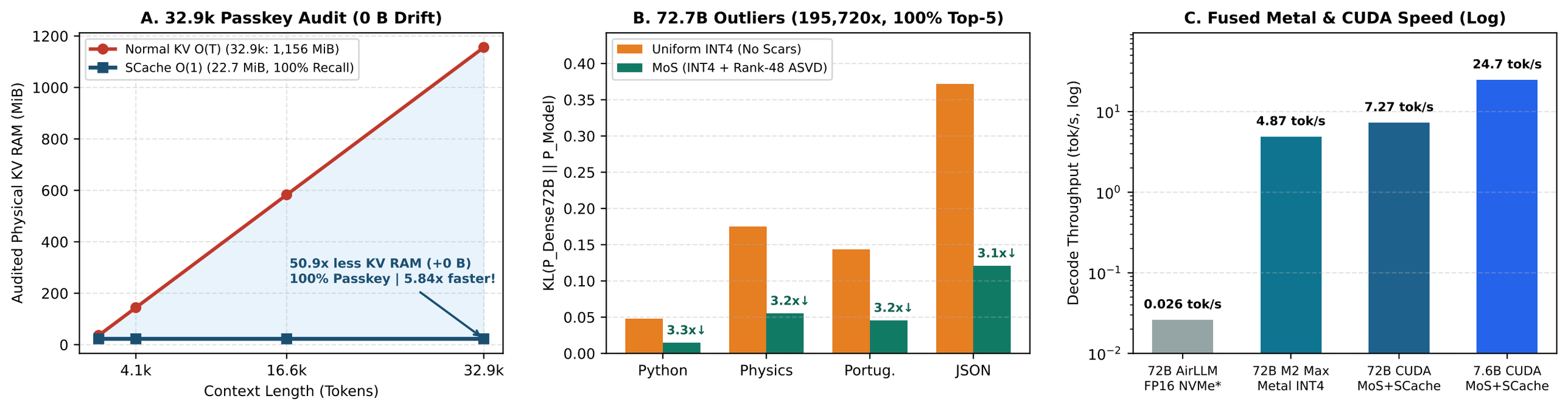
A. 32.9k Passkey Audit (0 B Drift)
Normal KV O(T) (32.9k: 1,156 MiB)
SCache O(1) (22.7 MiB, 100% Recall)
50.9x less KV RAM (+0 B)
100% Passkey | 5.84x faster!
Audited Physical KV RAM (MiB)
Context Length (Tokens)
4.1k
16.6k
32.9k
B. 72.7B Outliers (195,720x, 100% Top-5)
Uniform INT4 (No Scars)
MoS (INT4 + Rank-48 ASVD)
KL(P_Dense72B || P_Model)
3.3x↓
3.2x↓
3.2x↓
3.1x↓
Python
Physics
Portug.
JSON
C. Fused Metal & CUDA Speed (Log)
Decode Throughput (tok/s, log)
0.026 tok/s
4.87 tok/s
7.27 tok/s
24.7 tok/s
72B AirLLM FP16 NVMe*
72B M2 Max Metal INT4
72B CUDA MoS+SCache
7.6B CUDA MoS+SCache